

Theme 6

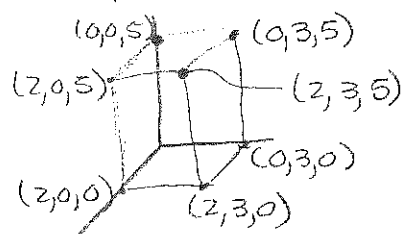
Unit 6.1

Ex 12.1 p796

3. $C(2, 4, 6)$ has x -value with smallest absolute value so it is closest to yz -plane.

$A(-4, 0, -1)$ lies in the xz -plane since $y=0$.

4. Projection of $(2, 3, 5)$ on
 xy -plane is $(2, 3, 0)$
 yz -plane is $(0, 3, 5)$
 xz -plane is $(2, 0, 5)$



length of diagonal - distance between origin and $(2, 3, 5)$

$$\sqrt{(2-0)^2 + (3-0)^2 + (5-0)^2} = \sqrt{38}$$

$$1. |PQ| = \sqrt{16+4+16} = 6$$

$$|QR| = \sqrt{36+4+0} = 2\sqrt{10}$$

$$|RP| = \sqrt{4+16+16} = 6$$

$$2. |PQ| = \sqrt{4+4+1} = 3$$

$$|QR| = \sqrt{0+36+9} = 3\sqrt{5}$$

$$|RP| = \sqrt{4+16+16} = 6$$

3. Equation of sphere with centre $(1, -4, 3)$ and $r=5$ is

$$(x-1)^2 + (y+4)^2 + (z-3)^2 = 5^2$$

$$\Rightarrow (x-1)^2 + (y+4)^2 + (z-3)^2 = 25$$

Intersection with xz -plane.

where $y=0$ therefore we have

$$(x-1)^2 + 4^2 + (z-3)^2 = 25$$

$$\Rightarrow (x-1)^2 + (z-3)^2 = 9, y=0$$

which is a circle in xz -plane with centre $(1, 0, 3)$ and radius 3.

15. Radius of sphere is distance between $(4, 3, -1)$ and $(3, 8, 1)$ so

$$r = \sqrt{(3-4)^2 + (8-3)^2 + (1-(-1))^2} = \sqrt{30}$$

So equation of sphere;

$$(x-3)^2 + (y-8)^2 + (z-1)^2 = 30$$

$$17. x^2 + y^2 + z^2 - 2x - 4y + 8z = 15$$

$$(x^2 - 2x + 1) + (y^2 - 4y + 4) +$$

$$(z^2 + 8z + 16) = 15 + 1 + 4 + 16$$

$$\therefore (x-1)^2 + (y-2)^2 + (z+4)^2 = 36$$

Sphere with centre $(1, 2, -4)$ and $r=6$.

nit 6.2

x 12.2 p805

Vector is equal if they have the same length and direction.

$$\overline{AB} = \overline{DC}, \overline{DA} = \overline{CB}, \overline{DE} = \overline{EB}, \overline{EA} = \overline{CE}.$$

$$(a) \overline{PQ} + \overline{QR} = \overline{PR}$$

initial point terminal point

$$(b) \overline{RP} + \overline{PS} = \overline{RS}$$

$$(c) \overline{QS} - \overline{PS} = \overline{QS} + \overline{SP} = \overline{QP}$$

$$(d) \overline{RS} + \overline{SP} + \overline{PQ} = (\overline{RS} + \overline{SP}) + \overline{PQ} = \overline{RP} + \overline{PQ} = \overline{RQ}$$

$$\overline{QR} = 2d$$

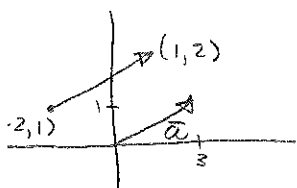
$$\overline{a} + 2\overline{d} = \overline{b} \Rightarrow 2\overline{d} = \overline{b} - \overline{a} \Rightarrow \overline{d} = \frac{\overline{b} - \overline{a}}{2} \text{ and}$$

$$\overline{c} + \overline{d} = \overline{b} \text{ so } \overline{d} = \overline{b} - \overline{c}$$

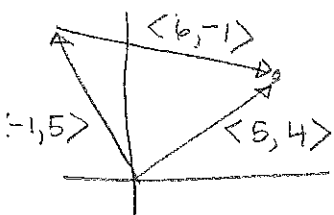
$$\Rightarrow \overline{b} - \overline{c} = \frac{1}{2}\overline{b} - \frac{1}{2}\overline{a}$$

$$\Rightarrow \overline{c} = \frac{1}{2}\overline{a} + \frac{1}{2}\overline{b}$$

$$\overline{a} = \langle 1 - (-2), 2 - 1 \rangle = \langle 3, 1 \rangle$$



$$\langle -1, 5 \rangle + \langle 6, -1 \rangle = \langle 5, 4 \rangle$$



$$\overline{a} + \overline{b} = \langle -3, 4 \rangle + \langle 9, -1 \rangle$$

$$= \langle 6, 3 \rangle$$

$$4\overline{a} + 2\overline{b} = 4\langle -3, 4 \rangle + 2\langle 9, -1 \rangle$$

$$= \langle 6, 14 \rangle$$

$$|a| = \sqrt{(-3)^2 + 4^2} = 5$$

$$|a-b| = | \langle -12, 5 \rangle | = 13.$$

$$21. \overline{a} = 4\overline{i} - 3\overline{j} + 2\overline{k}$$

$$\overline{b} = 2\overline{i} - 4\overline{k}$$

$$\overline{a} + \overline{b} = 6\overline{i} - 3\overline{j} - 2\overline{k}$$

$$4\overline{a} + 2\overline{b} = 16\overline{i} - 12\overline{j} + 8\overline{k} + 4\overline{i} - 8\overline{k} = 20\overline{i} - 12\overline{j}$$

$$|\overline{a}| = \sqrt{4^2 + (-3)^2 + 2^2} = \sqrt{29}$$

$$|\overline{a}-\overline{b}| = \sqrt{2^2 + (-3)^2 + 6^2} = 7$$

22.

$$\overline{a} + \overline{b} = \langle 13, -1, -3 \rangle$$

$$4\overline{a} + \overline{b} = \langle 42, 0, -14 \rangle$$

$$|\overline{a}| = \sqrt{8^2 + 1^2 + (-4)^2} = 9$$

$$|\overline{a}-\overline{b}| = | \langle 3, 3, -5 \rangle | = \sqrt{43}$$

23. $\langle 6, -2 \rangle$ has length

$| \langle 6, -2 \rangle | = \sqrt{40}$ so unit vector with same direction is

$$\frac{1}{\sqrt{40}} \langle 6, -2 \rangle = \left\langle \frac{3}{\sqrt{10}}, \frac{-1}{\sqrt{10}} \right\rangle$$

26. $| \langle 6, 2, -3 \rangle | = 7$ so unit

vector in direction of $\langle 6, 2, -3 \rangle$

is $\overline{u} = \frac{1}{7} \langle 6, 2, -3 \rangle$ and

therefore of length 4 will

$$\text{be } 4\overline{u} = \frac{4}{7} \langle 6, 2, -3 \rangle$$

$$= \left\langle \frac{24}{7}, \frac{8}{7}, -\frac{12}{7} \right\rangle$$

29. x-component of $\overline{v}$:

$$v_1 = |\overline{v}| \cos\left(\frac{\pi}{3}\right)$$

$$= 4 \cdot \frac{1}{2} = 2$$

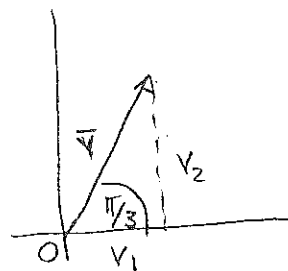
y-component of $\overline{v}$:

$$v_2 = |\overline{v}| \sin\left(\frac{\pi}{3}\right)$$

$$= 4 \cdot \frac{\sqrt{3}}{2} = 2\sqrt{3} \text{ Thus}$$

$$\overline{v} = \langle v_1, v_2 \rangle = \langle 2, 2\sqrt{3} \rangle$$

(2)



Unit 6.3

Ex 12.3 p812

1. (a) $\vec{a} \cdot \vec{b}$ is a scalar and dot product is defined for vectors so $(\vec{a} \cdot \vec{b}) \cdot \vec{c}$ has no meaning

(b) $(\vec{a} \cdot \vec{b}) \vec{c}$ is a scalar $\times$ vector - it has meaning.

(c) $|\vec{a}|$ and $\vec{b} \cdot \vec{c}$ are scalars so $|\vec{a}|(\vec{b} \cdot \vec{c})$ has meaning

(d) $\vec{a}$ and $\vec{b} + \vec{c}$ are vectors so the dot product $\vec{a} \cdot (\vec{b} + \vec{c})$ has meaning.

e) $\vec{a} \cdot \vec{b}$ is a scalar and $\vec{c}$ is a vector so $\vec{a} \cdot \vec{b} + \vec{c}$ is not possible.

f) $|\vec{a}|$ is a scalar - dot product defined only for vectors so $|\vec{a}| \cdot (\vec{b} + \vec{c})$ has no meaning.

$$\begin{aligned} 3. \vec{a} \cdot \vec{b} &= \langle 1.5, 0.4 \rangle \cdot \langle -4, 6 \rangle \\ &= (1.5)(-4) + (0.4)(6) \\ &= -3.6 \end{aligned}$$

$$\begin{aligned} 4. \vec{a} \cdot \vec{b} &= \langle 6, 4, 1 \rangle \cdot \langle 2, 5, -9 \rangle \\ &= (6)(2) + (4)(5) + (1)(-9) \\ &= 23 \end{aligned}$$

$$\begin{aligned} 7. \vec{a} \cdot \vec{b} &= (\vec{i} - 2\vec{j} + 3\vec{k}) \cdot (5\vec{i} + 9\vec{k}) \\ &= (1)(5) + (-2)(0) + (3)(9) \\ &= 32 \end{aligned}$$

11. $\vec{u}$, $\vec{v}$ and $\vec{w}$ are all unit vectors. - Δ equilateral
Angle between $\vec{u}$ and $\vec{v}$ is $\frac{\pi}{3}$ (60°) $\Rightarrow \vec{u} \cdot \vec{v} = |\vec{u}||\vec{v}|\cos\frac{\pi}{3}$
 $= (1)(1)\frac{1}{2} = \frac{1}{2}$

Move $\vec{w}$ to same initial point as $\vec{u}$ then the angle between $\vec{w}$ and $\vec{u}$ is $\frac{2\pi}{3}$ (120°) so

$$\begin{aligned} \vec{u} \cdot \vec{w} &= |\vec{u}||\vec{w}|\cos\frac{2\pi}{3} \\ &= (1)(1)(-\frac{1}{2}) = -\frac{1}{2} \end{aligned}$$

12. $\vec{u}$ is a unit vector so $\vec{w}$ is also a unit vector and $|\vec{v}| = |\vec{u}|\cos\frac{\pi}{4}$ ($\therefore$)

$$= (1)(\frac{1}{\sqrt{2}}) = \frac{1}{\sqrt{2}} \text{ so}$$

$$\begin{aligned} \vec{u} \cdot \vec{v} &= |\vec{u}||\vec{v}|\cos\frac{\pi}{4} \\ &= (1)(\frac{1}{\sqrt{2}})(\frac{1}{\sqrt{2}}) = \frac{1}{2} \text{ and} \end{aligned}$$

$\vec{u}$ and $\vec{w}$ orthogonal $\Rightarrow \vec{u} \cdot \vec{w} = 0$

$$\begin{aligned} 15. \vec{a} &= \langle 4, 3 \rangle & \vec{b} &= \langle 2, -1 \rangle \\ |\vec{a}| &= 5 & |\vec{b}| &= \sqrt{5} \end{aligned}$$

$$\text{and } \vec{a} \cdot \vec{b} = 5$$

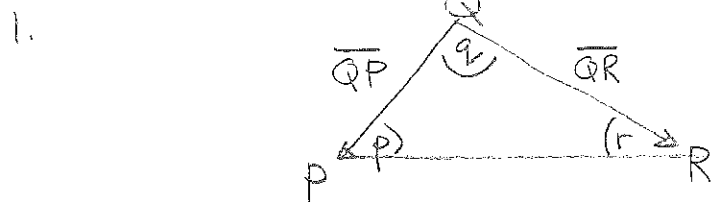
$$\begin{aligned} \cos \theta &= \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{5}{(5)\sqrt{5}} \\ &= \frac{1}{\sqrt{5}} \end{aligned}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

$$\begin{aligned} 19. |\vec{a}| &= \sqrt{26} & |\vec{b}| &= \sqrt{5} \\ \vec{a} \cdot \vec{b} &= 7 \end{aligned}$$

$$\text{Then } \cos \theta = \frac{7}{\sqrt{26} \cdot \sqrt{5}} = \frac{7}{\sqrt{130}}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{7}{\sqrt{130}}\right) \quad (3)$$



is the angle between $\overline{PQ}$ and $\overline{PR}$, etc.

$$\cos p = \frac{\overline{PQ} \cdot \overline{PR}}{|\overline{PQ}| |\overline{PR}|} = \frac{10}{\sqrt{221}}$$

$$\Rightarrow p = \cos^{-1}\left(\frac{10}{\sqrt{221}}\right) \approx 48^\circ$$

imilarly $\cos q = \frac{\overline{QP} \cdot \overline{QR}}{|\overline{QP}| |\overline{QR}|} = \frac{3}{\sqrt{130}}$

$$\Rightarrow q = \cos^{-1}\left(\frac{3}{\sqrt{130}}\right) \approx 75^\circ$$

$$\Rightarrow r \approx 180^\circ - (48^\circ + 75^\circ) = 57^\circ$$

3. (a) $\overline{a} \cdot \overline{b} = 0 \Rightarrow \overline{a}$ and

$\overline{b}$ are orthogonal not paral.

(b) $\overline{a} \cdot \overline{b} = -3 \neq 0$ and $\overline{a}$ is not a scalar multiple of $\overline{b}$, so $\overline{a}$ and $\overline{b}$ neither $\perp$ or $\parallel$.

(c) $\overline{a} \cdot \overline{b} = -168 \neq 0$ and

$\overline{a} = -\frac{4}{3}\overline{b}$ so $\overline{a}$ and $\overline{b}$ is parallel

(d) $\overline{a} \cdot \overline{b} = 0$ so $\overline{a}$ and $\overline{b}$ are orthogonal.

5. $\overline{QP} = \langle -1, -3, 2 \rangle$

$\overline{QR} = \langle 4, -2, -1 \rangle$ and

$\overline{QP} \cdot \overline{QR} = 0 \Rightarrow \overline{QP} \perp \overline{QR}$

so angle of triangle with vertex Q is a right angle.

27. $\overline{a} = a_1\overline{i} + a_2\overline{j} + a_3\overline{k}$

Say $\overline{a}$ is orthogonal to $\overline{i} + \overline{j}$ and $\overline{i} + \overline{k}$. then

$$\overline{a} \cdot (\overline{i} + \overline{j}) = 0 \Rightarrow$$

$$\langle a_1, a_2, a_3 \rangle \cdot \langle 1, 1, 0 \rangle$$

$$= a_1 + a_2 = 0 \text{ and}$$

$$\overline{a} \cdot (\overline{i} + \overline{k}) = 0 \Rightarrow$$

$$a_1 + a_3 = 0$$

So $a_1 = -a_2 = -a_3$ and

$\overline{a}$ is a unit vector so

$$a_1^2 + a_2^2 + a_3^2 = 3a_1^2 = 1$$

$$\Rightarrow a_1 = \pm \frac{1}{\sqrt{3}}$$

$$\Rightarrow \overline{a} = \frac{1}{\sqrt{3}}\overline{i} - \frac{1}{\sqrt{3}}\overline{j} - \frac{1}{\sqrt{3}}\overline{k} \text{ and}$$

$$\overline{a} = -\frac{1}{\sqrt{3}}\overline{i} + \frac{1}{\sqrt{3}}\overline{j} + \frac{1}{\sqrt{3}}\overline{k} \text{ are}$$

two such vectors.

29. $y = 2x - 3$ has slope 2 so

a vector parallel to the line

is $\langle 1, 2 \rangle$. $y = -3x + 7$ has

slope -3, so vector parallel

is $\langle 1, -3 \rangle$. The angle between

these vectors is the same

as the angle between the

given lines.

$$\cos \theta = \frac{\langle 1, 2 \rangle \cdot \langle 1, -3 \rangle}{|\langle 1, 2 \rangle| |\langle 1, -3 \rangle|}$$

$$= \frac{1 - 6}{\sqrt{5}\sqrt{10}} = \frac{-5}{5\sqrt{2}} = -\frac{1}{\sqrt{2}}$$

$$\Rightarrow \theta = \frac{3\pi}{4} \text{ so the}$$

acute angle between the line are $\frac{\pi}{4}$

(4)

33. Since

$|\langle 2, 1, 2 \rangle| = 3$ we have

$$\cos \alpha = \frac{2}{3} \Rightarrow \alpha = \cos^{-1} \frac{2}{3} \approx 48^\circ$$

$$\cos \beta = \frac{1}{3} \Rightarrow \beta = \cos^{-1} \frac{1}{3} \approx 71^\circ$$

$$\cos \gamma = \frac{2}{3} \Rightarrow \gamma = \cos^{-1} \frac{2}{3} \approx 48^\circ$$

36. Since

$$|\frac{1}{2}\bar{i} + \bar{j} + \bar{k}| = \sqrt{\frac{1}{4} + 1 + 1} = \frac{3}{2}$$

$$\cos \alpha = \frac{1/2}{3/2} = \frac{1}{3} \Rightarrow \alpha = \cos^{-1}(\frac{1}{3})$$

$$\cos \beta = \frac{2}{3} \approx 71^\circ$$

$$\cos \gamma = \frac{2}{3} \quad \beta = \gamma = \cos^{-1}(\frac{2}{3}) \approx 48^\circ$$

38. Since

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\begin{aligned} \cos^2 \gamma &= 1 - \cos^2 \alpha - \cos^2 \beta \\ &= 1 - \cos^2(\frac{\pi}{4}) - \cos^2(\frac{\pi}{3}) \\ &= 1 - (\frac{1}{\sqrt{2}})^2 - (\frac{1}{2})^2 \\ &= \frac{1}{4} \end{aligned}$$

$$\Rightarrow \cos \gamma = \pm \frac{1}{2} \text{ and}$$

$$\gamma = \frac{\pi}{3} \text{ or } \gamma = \frac{2\pi}{3}$$

39. $|\bar{a}| = 5$ and so

scalar projection of $\bar{b}$ onto

$\bar{a}$ is $\text{comp}_{\bar{a}} \bar{b} = \frac{\bar{a} \cdot \bar{b}}{|\bar{a}|}$

$$= \frac{(4)(6) + (3)(7)}{5}$$

$$= 9$$

and the vector projection of $\bar{b}$ onto $\bar{a}$ is

$$\text{proj}_{\bar{a}} \bar{b} = \left(\frac{\bar{a} \cdot \bar{b}}{|\bar{a}|} \right) \frac{\bar{a}}{|\bar{a}|} = (9 \cdot \frac{1}{5}) \langle 4, 3 \rangle$$

$$= \langle \frac{36}{5}, \frac{27}{5} \rangle$$

41. $|\bar{a}| = 9$ so

$$\text{comp}_{\bar{a}} \bar{b} = \frac{\bar{a} \cdot \bar{b}}{|\bar{a}|} = \frac{1}{9} \text{ and}$$

$$\begin{aligned} \text{proj}_{\bar{a}} \bar{b} &= \frac{1}{9} \left(\frac{\bar{a}}{|\bar{a}|} \right) = \frac{1}{9} \cdot \frac{1}{9} \langle 4, 7, -4 \rangle \\ &= \langle \frac{4}{81}, \frac{7}{81}, -\frac{4}{81} \rangle \end{aligned}$$

43. $|\bar{a}| = \sqrt{19}$

$$\text{comp}_{\bar{a}} \bar{b} = \frac{\bar{a} \cdot \bar{b}}{|\bar{a}|} = -\frac{7}{\sqrt{19}}$$

$$\begin{aligned} \text{proj}_{\bar{a}} \bar{b} &= \left(-\frac{7}{\sqrt{19}} \right) \frac{\bar{a}}{|\bar{a}|} \\ &= -\frac{7}{\sqrt{19}} \cdot \frac{1}{\sqrt{19}} \langle 3, -3, 1 \rangle \\ &= -\frac{21}{19} \bar{i} + \frac{21}{19} \bar{j} - \frac{7}{19} \bar{k} \end{aligned}$$

45. $(\text{orth}_{\bar{a}} \bar{b}) \cdot \bar{a}$

$$\begin{aligned} &= (\bar{b} - \text{proj}_{\bar{a}} \bar{b}) \cdot \bar{a} \\ &= \bar{b} \cdot \bar{a} - (\text{proj}_{\bar{a}} \bar{b}) \cdot \bar{a} \\ &= \bar{b} \cdot \bar{a} - \frac{\bar{a} \cdot \bar{b}}{|\bar{a}|^2} \bar{a} \cdot \bar{a} \\ &= \bar{b} \cdot \bar{a} - \frac{\bar{a} \cdot \bar{b}}{|\bar{a}|^2} \cdot |\bar{a}|^2 \\ &= \bar{b} \cdot \bar{a} - \bar{a} \cdot \bar{b} = 0 \end{aligned}$$

So $\text{orth}_{\bar{a}} \bar{b}$ and $\bar{a}$ are orth.

46. $\text{orth}_{\bar{a}} \bar{b} = \bar{b} - \text{proj}_{\bar{a}} \bar{b}$

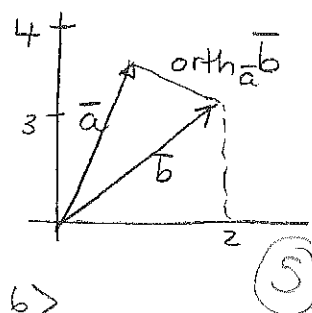
$$\begin{aligned} &= \langle 2, 3 \rangle - \frac{28}{37} \langle -1, 6 \rangle \\ &= \langle \dots \rangle \end{aligned}$$

40.*

$$|\bar{a}| = \sqrt{37}$$

$$\begin{aligned} \text{proj}_{\bar{a}} \bar{b} &= \left(\frac{\bar{a} \cdot \bar{b}}{|\bar{a}|} \right) \frac{\bar{a}}{|\bar{a}|} \\ &= \left(\frac{28}{\sqrt{37}} \right) \frac{1}{\sqrt{37}} \langle -1, 6 \rangle \end{aligned}$$

$$= \frac{28}{37} \langle -1, 6 \rangle$$



(5)

nit 6.4

x 12.4 p821

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 0 \\ 1 & 0 & 5 \end{vmatrix}$$

$$= 15\vec{i} - 10\vec{j} - 3\vec{k}$$

$$(\vec{a} \times \vec{b}) \cdot \vec{a} = \langle 15, -10, -3 \rangle \cdot \langle 2, 3, 0 \rangle$$

$$= 0 \quad \text{and}$$

$$(\vec{a} \times \vec{b}) \cdot \vec{b} = \langle 15, -10, -3 \rangle \cdot \langle 1, 0, 5 \rangle$$

$$= 0$$

$\vec{a} \times \vec{b}$ orthogonal to both $\vec{a}$ and $\vec{b}$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 2 & -4 \\ -1 & 3 & 1 \end{vmatrix}$$

$$= 14\vec{i} + 4\vec{j} + 2\vec{k}$$

$$(\vec{a} \times \vec{b}) \cdot \vec{a} = 0 \quad \text{and} \quad (\vec{a} \times \vec{b}) \cdot \vec{b} = 0$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ t & 1 & t \\ t^2 & t^2 & 1 \end{vmatrix}$$

$$= (1-t)\vec{i} + (t^3-t^2)\vec{k}$$

$$(\vec{a} \times \vec{b}) \cdot \vec{a} = (\vec{a} \times \vec{b}) \cdot \vec{b} = 0$$

(a) $\vec{b} \times \vec{c}$ is a vector so

$\vec{a}(\vec{b} \times \vec{c})$ is meaningful

(b) $\vec{b} \cdot \vec{c}$ is a scalar so

$\vec{a} \times (\vec{b} \cdot \vec{c})$ is meaningless

(c) $\vec{b} \times \vec{c}$ is a vector so

$\vec{a} \times (\vec{b} \times \vec{c})$ is meaningful.

(d) $\vec{b} \cdot \vec{c}$ is a scalar so

$\vec{a} \cdot (\vec{b} \cdot \vec{c})$ is meaningless.

(e) $(\vec{a} \cdot \vec{b})$ and $(\vec{c} \cdot \vec{d})$ are both scalars so cross product of these two are meaningless.

(f) $\vec{a} \times \vec{b}$ and $\vec{c} \times \vec{d}$ are vectors so their dot product is meaningful.

19. $\langle 3, 2, 1 \rangle \times \langle -1, 1, 0 \rangle$

$$= -\vec{i} - \vec{j} + 5\vec{k}$$

So unit vector orthogonal to both are

$$\pm \frac{\langle -1, -1, 5 \rangle}{\sqrt{1+1+25}} = \pm \frac{\langle -1, -1, 5 \rangle}{3\sqrt{3}}$$

20. $\langle +\vec{j} - \vec{k} \rangle \times \langle \vec{i} + \vec{j} \rangle$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 1 & -1 \\ 1 & 1 & 0 \end{vmatrix} = \langle 1, -1, -1 \rangle$$

Two ^{unit} vectors orthogonal to given vectors are

$$\pm \frac{\langle 1, -1, -1 \rangle}{\sqrt{1+1+1}} = \pm \langle \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \rangle$$

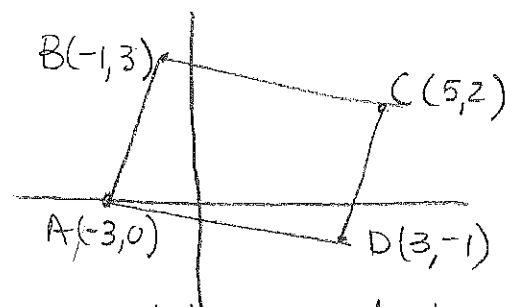
$$= \frac{1}{\sqrt{3}}\vec{i} - \frac{1}{\sqrt{3}}\vec{j} - \frac{1}{\sqrt{3}}\vec{k} \quad \text{or}$$

$$-\frac{1}{\sqrt{3}}\vec{i} + \frac{1}{\sqrt{3}}\vec{j} + \frac{1}{\sqrt{3}}\vec{k}$$

27.

$$\vec{AB} = \langle 2, 3 \rangle$$

$$\vec{AD} = \langle 6, -1 \rangle$$



Area of parallelogram determined by these two vectors is the length of the cross product.

Consider $\vec{AB}$ as 3-dim vector $\langle 2, 3, 0 \rangle$ and $\vec{AD}$ as $\langle 6, -1, 0 \rangle$

(6)

$$\begin{aligned}\overline{AB} \times \overline{AD} &= \begin{vmatrix} \overline{i} & \overline{j} & \overline{k} \\ 2 & 3 & 0 \\ 6 & -1 & 0 \end{vmatrix} \\ &= -20\overline{k} \\ &= \langle 0, 0, -20 \rangle\end{aligned}$$

$$|\overline{AB} \times \overline{AD}| = \sqrt{20^2} = 20$$

Area of parallelogram is 20.

43. $|\overline{a} \times \overline{b}| = |\overline{a}| |\overline{b}| \sin \theta$ where θ the angle between $\overline{a}$ and $\overline{b}$ and $\overline{a} \cdot \overline{b} = |\overline{a}| |\overline{b}| \cos \theta$

$$\Rightarrow |\overline{a}| |\overline{b}| = \frac{\overline{a} \cdot \overline{b}}{\cos \theta} \text{ so}$$

$$\frac{|\overline{a} \times \overline{b}|}{\overline{a} \cdot \overline{b}} = \frac{\sin \theta}{\cos \theta} = \tan \theta$$

$$|\overline{a} \times \overline{b}| = |\langle 1, 2, 2 \rangle| = 3$$

$$\text{so } \tan \theta = \frac{3}{\sqrt{3}} = \sqrt{3}$$

$$\Rightarrow \theta = \frac{\pi}{3} (60^\circ)$$